

CHRISTOPHER CAPIZZUTO

SOFTWARE ENGINEER | FULL-STACK, MOBILE & APPLIED AI

Builds web and mobile products, computer-vision features, and autonomous gameplay agents. Creator of Dragapultist, used by 50+ users, and founder of Brightboard Chess.

Yonkers, NY | (347) 462-6803 | Chriszcodes@gmail.com

github.com/capisz | [linkedin.com/in/chriszcodes](https://www.linkedin.com/in/chriszcodes) | capizzuto-portfolio.vercel.app

U.S. citizen; authorized to work without sponsorship | Fluent in English and Spanish

EXPERIENCE & LEADERSHIP

Brightboard Chess | Founder

Current

- Founded an AI-focused chess education nonprofit to expand access for Black, Brown, and underserved students in New York City and beyond.
- Developing an instructional model that combines traditional chess teaching with AI-assisted learning and practice.
- **Finalist for the Andy Lerner Award for Chess Education Excellence.**

Independent Mobile & Full-Stack Developer

2021 - Present

- Build and maintain web and mobile products, owning frontend development, APIs, databases, deployment, and production debugging.
- Implement authentication, third-party API integrations, and resilient payment workflows, including OAuth token handling and idempotent Stripe webhooks.

Northeast STEM Starter Academy (NSSA) | Head of Chess

Mount Vernon, NY | April 2026 - Present

- Lead three weekly chess classes for elementary and middle school students at a STEM education nonprofit; designed and taught a three-week camp across three skill levels.
- Created the operating plan and \$37,968 budget for a proposed Saturday Chess Academy expansion, including skill placement, staffing, and student readiness criteria.

SparrowUp | Software Engineer

May 2025 - April 2026

- Built motion detection and tracking for a soccer training app, including arm and leg segmentation to analyze player movement.
- Translated motion-capture output into coaching feedback using seven biomechanical variables, including strike speed, rotation, knee bend, and plant-foot position, across four movement stages.
- Documented tracking edge cases, test scenarios, and measurement inconsistencies to support pre-release validation.

North American Dental Group | Accountant

Ridge Hill office, Yonkers, NY | 2019 - 2023

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Swift, SQL, HTML/CSS

Web & mobile: React, Next.js, React Native, Expo, SwiftUI, Electron, Tailwind CSS

Backend & AI: Node.js, Express, FastAPI, PyTorch, REST APIs, Claude API

Data: PostgreSQL/PostGIS, MongoDB, Firebase

Delivery & testing: Git, Docker, Kubernetes (local kind), GitHub Actions, Vercel, Playwright, Vitest, XCUITest, Firebase Emulator Suite

SELECTED ENGINEERING PROJECTS

Pokémon TCG AI | Autonomous Gameplay & Kaggle Research

Python, TypeScript, PyTorch, FastAPI, React | In development

- Achieved a **top-16 finish** in The Pokémon Company - PTCG AI Battle Challenge Strategy on Kaggle; currently participating in the AI Battle Challenge Playground competition.
- Built a Python gameplay agent that ranks simulator-provided legal actions, prioritizing board setup, evolution, energy attachment, and attacks.
- Developing a local research lab connecting autonomous self-play, policy training, saved checkpoints, and replay analysis; keeping experimental results separate from validated evaluation.

Dragapultist | Pokémon TCG Match Analyzer

Next.js, React, TypeScript, MongoDB, Electron | Live · GitHub

- Built and maintain a match-analysis app **used by 50+ users**, turning Pokémon TCG Live exports into game history, win rates, and matchup insights.
- Designed indexed MongoDB schemas and aggregation pipelines for match statistics, with an Electron desktop build supporting local persistence and automated data ingestion.

PrizeCheck | Pokémon TCG Practice Trainer

Next.js, TypeScript, Firebase, Docker, Kubernetes, GitHub Actions | Live · GitHub

- Built a deck-import practice tool that simulates opening hands and prize cards, then scores recall by accuracy and speed.
- Implemented Google sign-in, saved decks, ownership-enforced Firestore rules, and offline-tolerant session sync with stable IDs to prevent duplicate records.
- Containerized the app and exercised pod replacement and rollback on a local kind cluster; configured CI with unit, security-rules, and browser tests.

backstop.ai | Catcher Analytics & AI Analyst

Next.js, TypeScript, Python, FastAPI, PostgreSQL, Claude API | Live · GitHub

- Built catcher pitch-calling analysis using public baseball data and pitcher-specific alternatives, served through a Python/FastAPI backend.
- Integrated a streaming Claude analyst grounded in the displayed zone report, with pitch-count caching to avoid repeated calls during live polling.

FunkFit | Adaptive Nutrition App

React Native, Expo, TypeScript, HealthKit, Fly.io | GitHub

- Built iOS and web clients from a shared TypeScript codebase with offline persistence and protected third-party credentials behind a backend proxy.
- Developed nutrition targets that use measured wearable energy when available, track the source of each estimate, and adjust goals to weight trends.

Pidge | NYC Parking Intelligence

SwiftUI, MapKit, Node.js, TypeScript, PostgreSQL/PostGIS | GitHub

- Built native and web map clients backed by spatial queries and normalized NYC curb data, including signs, meters, and street geometry.
- Designed conservative recommendations that exclude unknown and no-parking curbs; added backend, browser, and iOS test coverage.

Additional work: settle.ai, a room planner with a Three.js editor and browser-based floor-plan extraction; CareCation, a travel platform integrating Amadeus search and Stripe payments.

EDUCATION

Georgia Institute of Technology | M.S. Computer Science, AI specialization (incoming)

Codesmith | Full-Stack Software Engineering, 2021

Baruch College (CUNY) | B.B.A., Digital Marketing, 2020